Day 2 afternoon — prediction and counts

Bayesian statistics for PhD researchers · Lecturer: Lu Cheng

Tuesday 10 November 2026 · Kroghstræde 3, room 4.110

Posterior prediction · Poisson counts with exposure · gamma–Poisson update · rate versus future count · how the distributions connect

Today

08:45	**Lecture A**	Proportions: beta–binomial updating · first look at Stan
09:45	**Exercise A**	Your own update in R, two priors, a grid check · question set
11:10	**Peer review A**	Partner interviews · plenary · everyone back at 11:10
11:55	**Lunch**	
12:30	**Lecture B**	Prediction · counts with exposure · map of distributions
13:30	**Exercise B**	Posterior predictive counts and one predictive plot · question set
15:00	**Peer review B**	Partner interviews · plenary · everyone back at 15:00
15:45	**Daily diary**	Mentimeter: 3 things learned, 3 not yet familiar

By the end of this lecture you can

1. **Predict** a future observation and say why it is wider than a plug-in guess
2. **Model** counts observed over different lengths of time (exposure)
3. **Update** a Gamma prior to a Gamma posterior
4. **Separate** uncertainty about a rate from uncertainty about a future count
5. **Follow** how one distribution leads to another (orientation only)

Posterior prediction: two kinds of uncertainty

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta) \cdot p(\theta \mid y)\, d\theta$$

1 We do not know θ

Posterior uncertainty — shrinks as data grow.

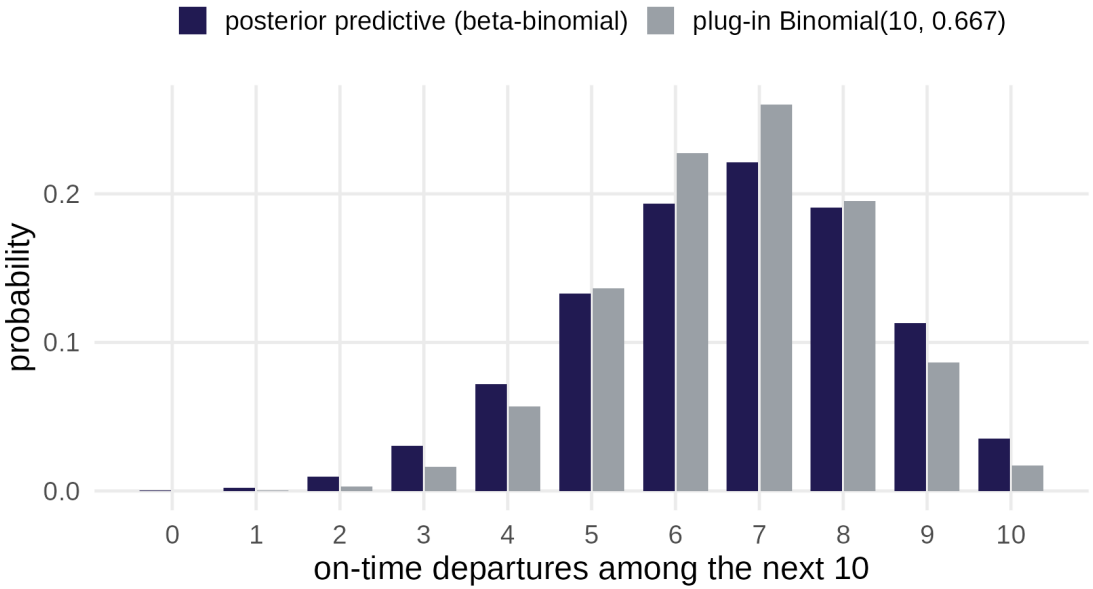
2 Departures vary even if θ were known

Sampling variation — never goes away.

```
theta_draw <- rbeta(10000, 16, 8)              # 1: draw θ from the posterior
y_next10   <- rbinom(10000, 10, theta_draw)     # 2: draw new data given θ
```

Simulate in two steps: first the parameter, then the data.

Predicting the next 10 departures



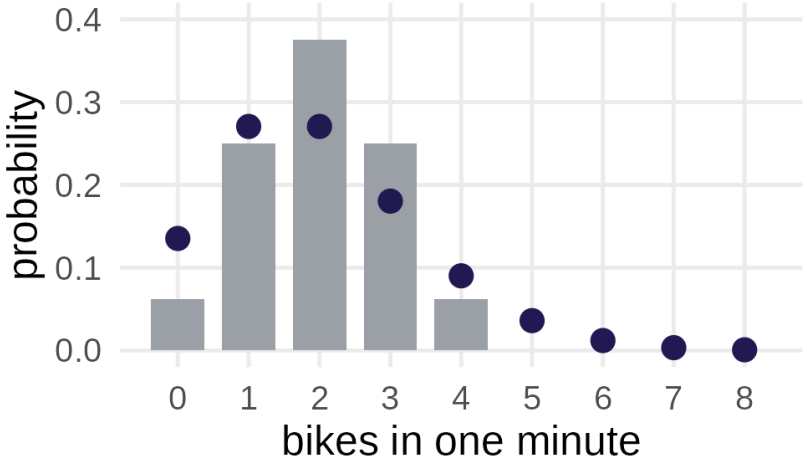
- Posterior predictive: SD 1.74 departures
- Plug-in Binomial(10, 0.667): SD 1.49 — too confident
- P(at least 8 on time): 0.34 vs 0.30
  - The exact predictive has a name: beta-binomial (a distribution, not the model)

Counting things: from binomial to Poisson

4 slices: p = 2/4 = 0.5 per slice



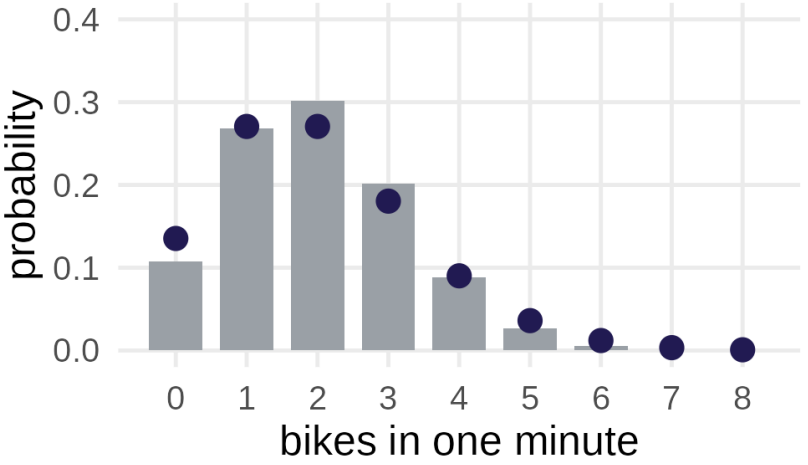
one minute · coral = a bike in that slice



10 slices: p = 2/10 = 0.2 per slice



one minute · coral = a bike in that slice



Grey bars: Binomial(m, 2/m) **Dark points:** Poisson(2). More slices → the bars move onto the points.

Exposure: counts over different lengths of time

session	minutes t_i	bikes y_i	per minute
1	10	23	2.30
2	15	31	2.07
3	5	9	1.80
4	20	44	2.20
5	10	18	1.80
total	**60**	**125**	**2.08**

$$y_i \mid \lambda \sim \text{Poisson}(\lambda \cdot t_i)$$

- λ = bikes per minute; t_i = minutes counted (the exposure).
- Longer sessions carry more information.
- The mean count, 25 per session, is not a rate.
- Exposure works because a sum of Poisson counts is again Poisson.

Invented data.

The gamma–Poisson update

$$p(\lambda \mid y) \propto \prod (\lambda t_i)^{y_i} e^{-\lambda t_i} \times \lambda^{a-1} e^{-b\lambda}$$
$$\propto \lambda^{a + \Sigma y_i - 1} e^{-(b + \Sigma t_i)\lambda}$$

$$\lambda \mid y \sim \text{Gamma}(a + \Sigma\, y_i,\ b + \Sigma\, t_i)$$

Gamma(a, b)

shape a, rate b · mean a/b

read: a events seen in b units of time

Add counts to a

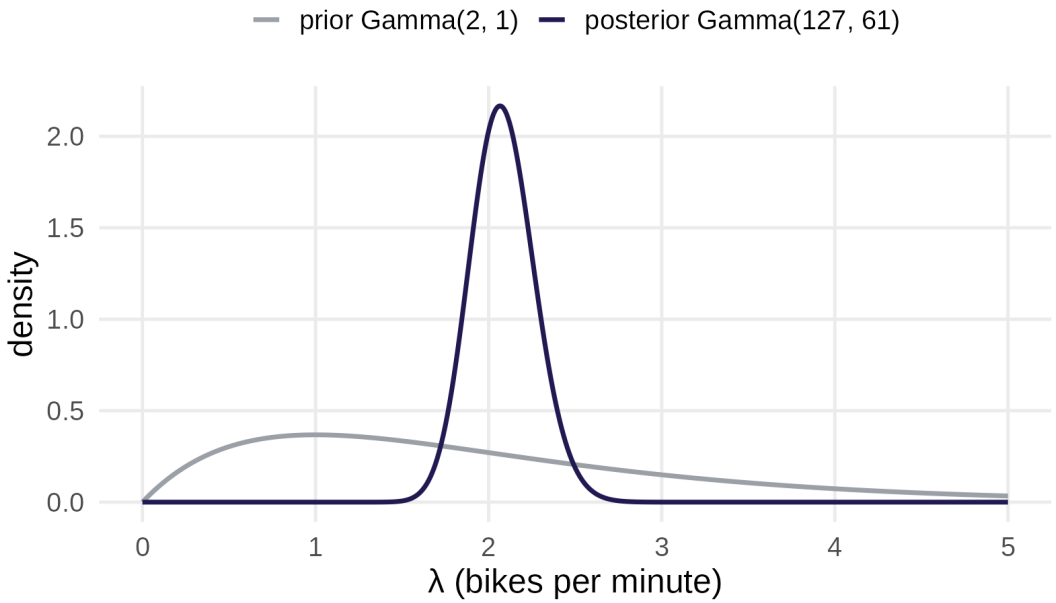
a = 2 → 2 + 125 = 127

Add exposure to b

b = 1 → 1 + 60 = 61

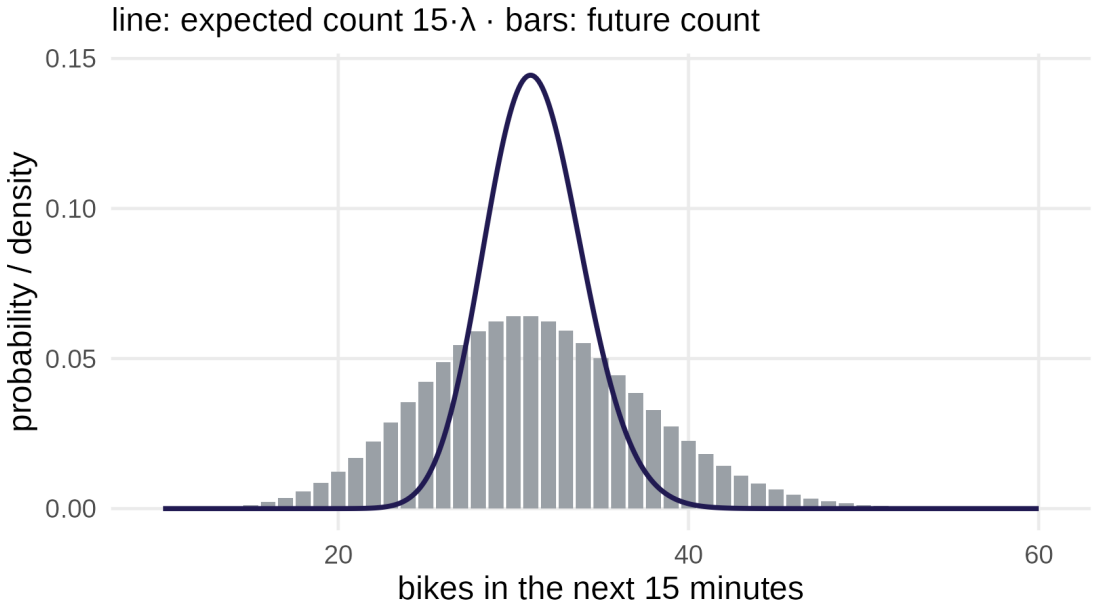
In R: dgamma(x, shape = a, rate = b) — check which parameterisation a function uses.

Bikes per minute: prior and posterior



- Prior Gamma(2, 1): mean 2, 90% interval 0.36 – 4.7
- Posterior Gamma(127, 61): mean 2.08, SD 0.18
- 90% interval 1.79 – 2.39 bikes per minute
- 60 minutes of counting outweigh the prior

A rate versus a future count



next 15 minutes	90% interval
expected count 15·λ	26.8 – 35.9
future count ỹ	21 – 42

The first is about λ; the second adds Poisson variation. A question about tomorrow's count needs the second.

The same model in Stan

```
data {
  int<lower=1> N;
  array[N] int<lower=0> y;
  vector<lower=0>[N] t;
  real<lower=0> a;  real<lower=0> b;
  real<lower=0> t_new;
}
parameters {
  real<lower=0> lambda;
}
model {
  lambda ~ gamma(a, b);
  y ~ poisson(lambda * t);     // exposure
}
generated quantities {
  int y_new = poisson_rng(lambda * t_new);
}
```

Exposure is one multiplication

lambda * t gives each session its own mean.

generated quantities

Runs once per draw: draw λ, then a new count — the two-step recipe from slide “two kinds of uncertainty”.

Demo 3 runs it and compares with the exact answer.

Worked example: one more session

New data

A second volunteer counts 7 bikes in 5 minutes.

Update Gamma(127, 61). Predict first: does the mean go up or down?

Longer horizon

Predict the next 60 minutes instead of 15.

Which interval grows faster: the expected count or the future count?

- 2 minutes with your neighbour, then we compute in R.

Worked example: answers

New data: 7 bikes in 5 minutes

Gamma(127 + 7, 61 + 5) = Gamma(134, 66)

mean 2.08 → 2.03 bikes per minute: down, because 7 / 5 = 1.4 per minute is below 2.08

90%: 1.79 – 2.39 → 1.75 – 2.33, slightly narrower

5 more minutes add little to 61.

Longer horizon (posterior Gamma(127, 61))

90% interval	next 15 min	next 60 min
expected count λt	26.8 – 35.9	107 – 144
future count	21 – 42	100 – 152

Widths: expected count 9 → 36 (×4, in proportion to t), future count 21 → 52 (×2.5). The expected count grows faster; the future count always stays wider.

Short horizon: Poisson noise dominates. Long horizon: uncertainty about λ dominates.

Relationships between distributions

Bernoulli(p) — one trial, two outcomes

- One departure: on time (1) with probability p, late (0) with probability 1 − p.
- Toss a coin; draw one marble from a bag.
- $P(X = x) = p^x\,(1 - p)^{1-x}$, $x \in \{0, 1\}$



Every distribution on the next slides is built from this one step.

Relationships between distributions

Binomial(n, p) — add up n independent Bernoulli(p) trials



$$Y = X_1 + X_2 + \ldots + X_n = 14$$

- X_i iid Bernoulli(p) ⇒ Y ~ Binomial(n, p): $P(Y = y) = C(n, y)\, p^y\, (1 - p)^{n-y}$
- Only the number of successes matters, not their order — the C(n, y) counts the orders.
- Binomial(n, p) with n = 1 ⇒ Bernoulli(p)

Relationships between distributions

Poisson(λ) — λ = 2 bikes per minute

one window: a bike or not, Bernoulli(p = λ/n)



divide 1 minute into n windows

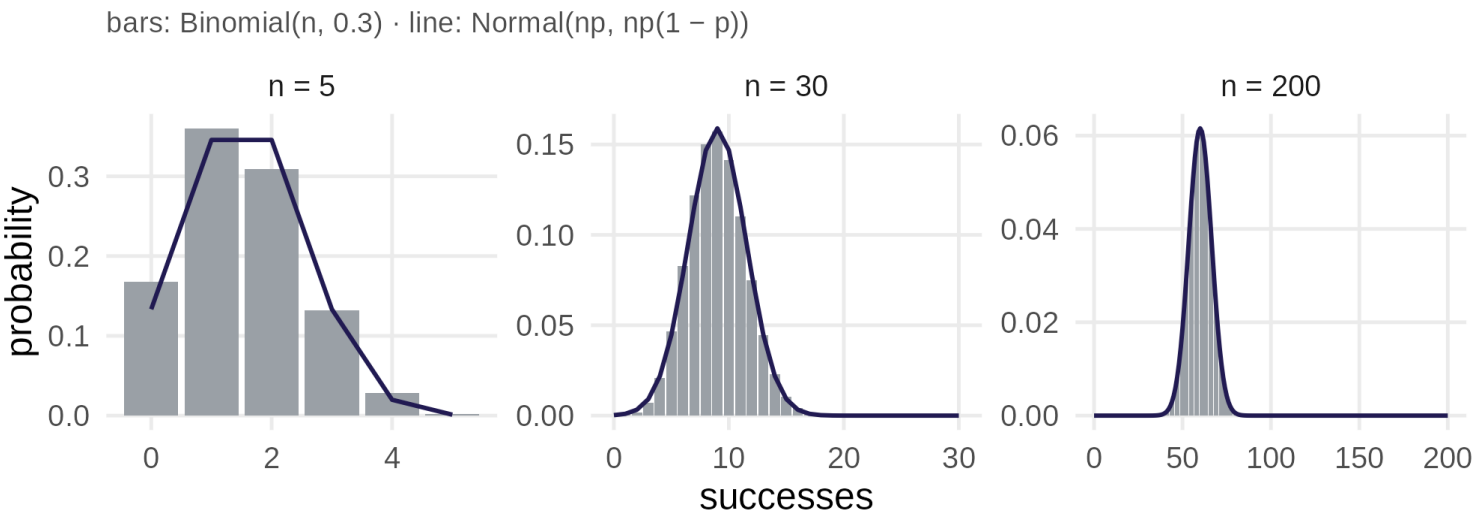
Binomial(n, p) ⇛ λ = np, n → ∞, p → 0 ⇛ Poisson(λ)

windows n	**4**	**20**	**100**	**1000**	**Poisson(2)**
P(2 bikes in the minute)	0.375	0.285	0.273	0.271	0.271

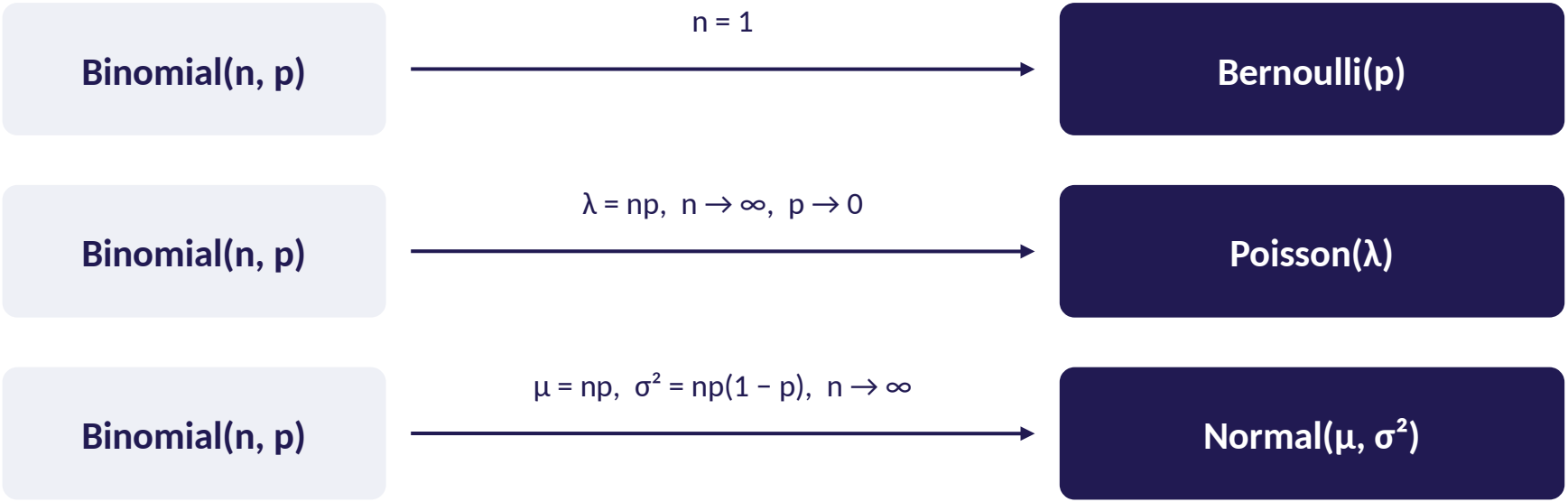
Relationships between distributions

Normal(μ, σ^2) — the central limit theorem

Binomial(n, p) ⇒ $\mu = np$, $\sigma^2 = np(1-p)$, $n \to \infty$ ⇒ Normal(μ, σ^2)



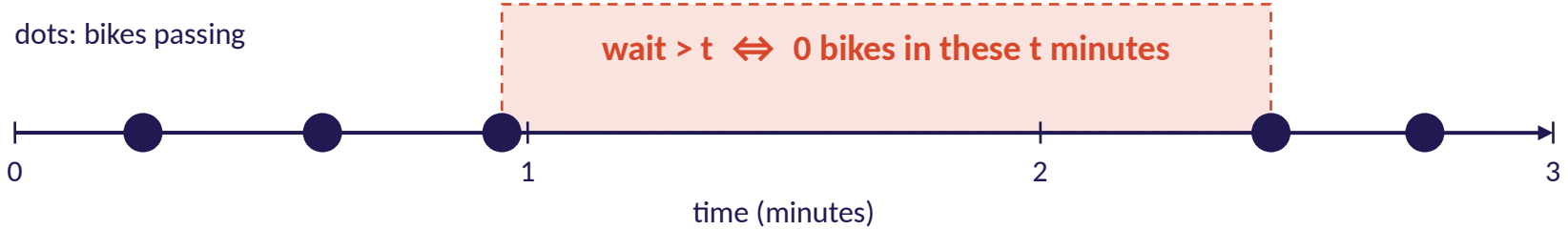
Relationships between distributions



Demo 2 checks each arrow by simulation. You do not need to memorise them.

Relationships between distributions

Exponential(λ) — the same bikes, timed instead of counted



Counting (Poisson)

Bikes in t minutes ~ Poisson(λt)

P(0 bikes) = $(\lambda t)^0 e^{-\lambda t} / 0! = e^{-\lambda t}$

⇔

Waiting (exponential)

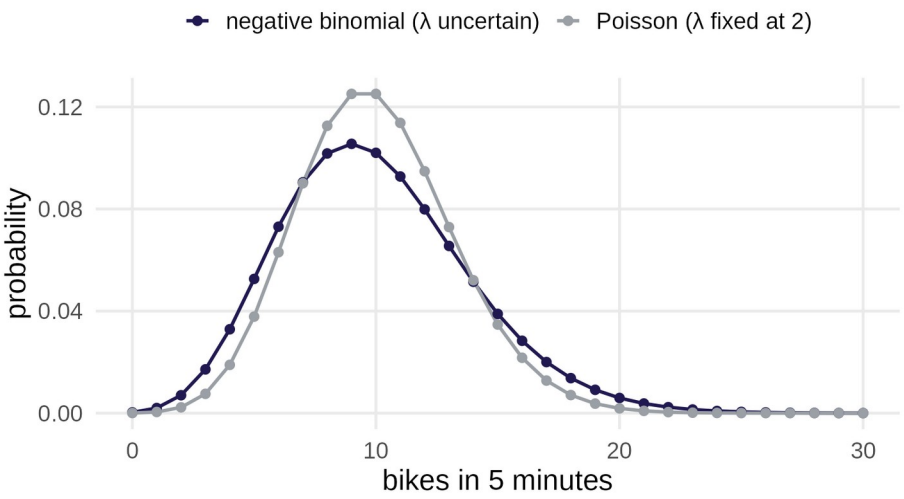
P(wait > t) = $e^{-\lambda t}$

density $\lambda e^{-\lambda t}$ · mean wait $1/\lambda$

λ = 2 per minute: mean gap 0.5 min · P(no bike for a whole minute) = $e^{-2} \approx 0.135$

Relationships between distributions

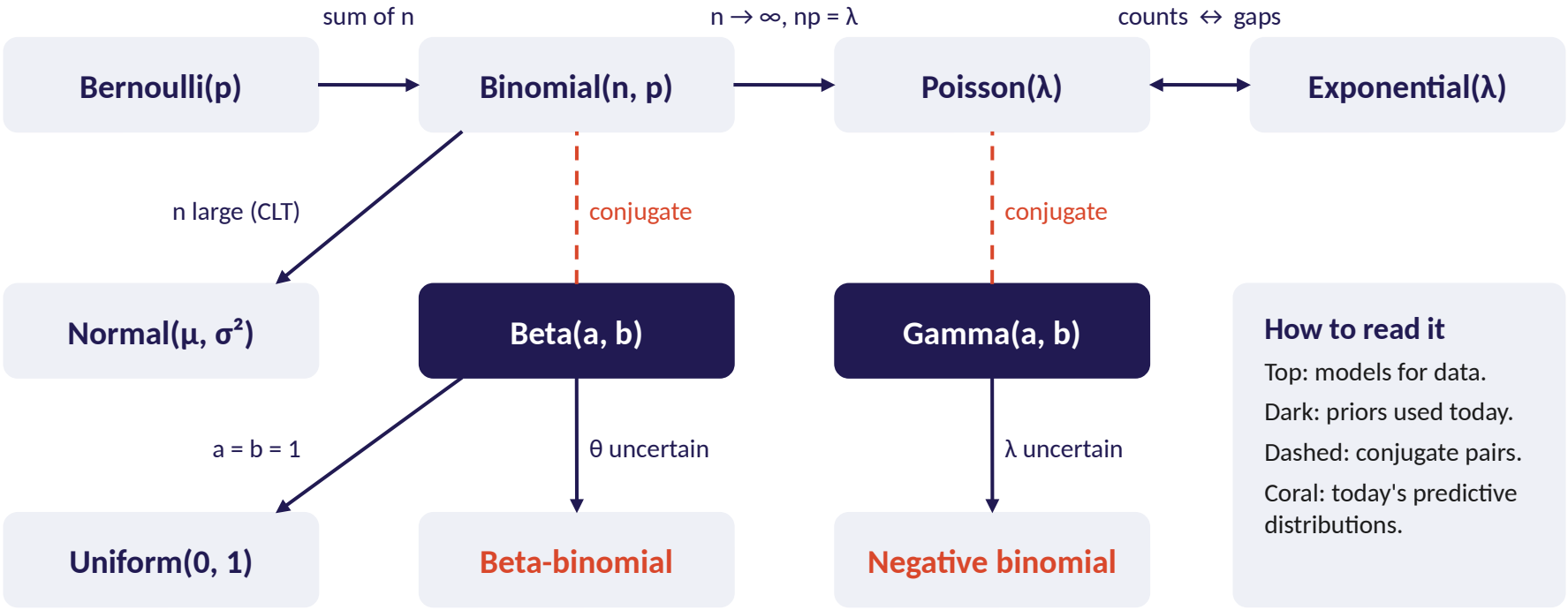
When the parameter itself is uncertain



data	parameter	together
Binomial(n, θ)	θ ~ Beta(a, b)	beta-binomial
Poisson(λ t)	λ ~ Gamma(a, b)	negative binomial

- Same mean, larger variance.
- With the posterior for θ or λ, these are today's two posterior predictive distributions.

Putting it together



For orientation only. Demo 2 checks every arrow by simulation.

Exercise B · 13:30 – 15:00

- Update a gamma prior with the supplied count data; mind the exposure.
- Simulate future counts from posterior rate draws.
- Make one prior or posterior predictive plot (ggplot2).
- Explain the difference between the rate and a future count.
- Write your group's question set: concept · computation · pitfall, with worked solutions.

Everyone back at

15:00

peer review B

Take a break when it suits your group.
New roles this block.

Peer review B · 15:00 – 15:45

20 min

Partner interviews

Round 1: odd group asks even group. Round 2: swap. Two questions each, to named members.

23 min

Plenary

Two groups drawn from those not yet drawn. Then the reporters' hardest open questions.

2 min

Wrap-up

Points that need longer go to tomorrow's follow-up.

Partner pairs: 1 ↔ 2 · 3 ↔ 4 · 5 ↔ 6 · 7 ↔ 8 · 9 ↔ 10

Running group scores for Days 1–2 are shown after today's session.

Daily diary · 15:45 – 16:00

menti.com · code: ____ ____

1. Three things I have learned today
2. Three things I am not yet very familiar with

To try after the course: the demos

Demo 1	**`demo-1-beta-binomial.ipynb`** The bus example: exact, sequential, prior sensitivity, grid, Stan, prediction
Demo 2	**`demo-2-distribution-connections.ipynb`** Every arrow of the map checked by simulation (R only)
Demo 3	**`demo-3-gamma-poisson-exposure.ipynb`** Bike counts with exposure: exact, Stan, rate vs future count, a quick check

Folder day-2/demos/ · open in JupyterLab with the R (bayes) kernel · Stan files are in models/ · each notebook ends with “Try this”.